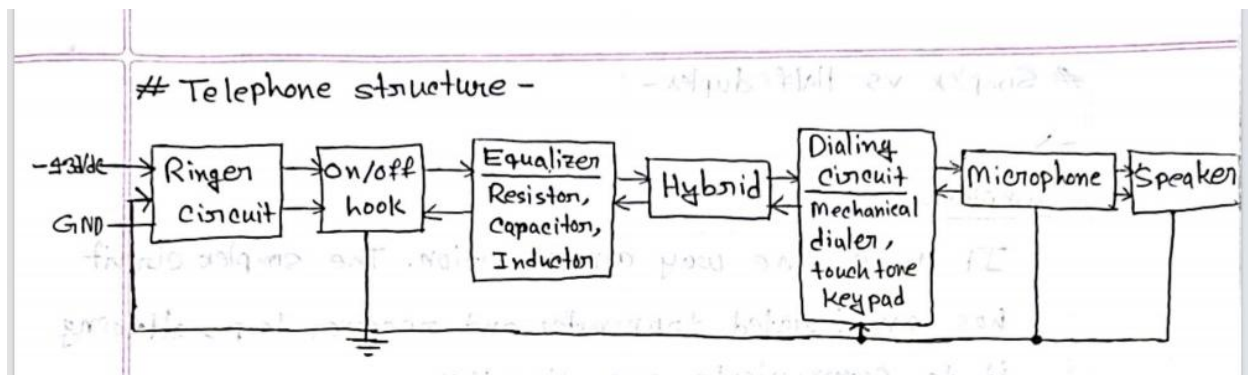




Ringer Circuit - Notifies the subscriber for incoming traffic

On/off hook - Mechanically connects the set.

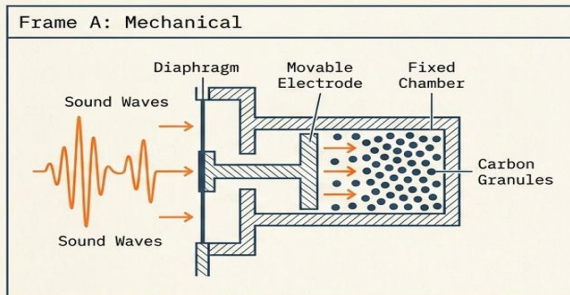
Idle On hook = Open circuit

In use off hook = Close circuit

Equalizer - Regulates amplitude and frequency response of the speech signal using passive component.

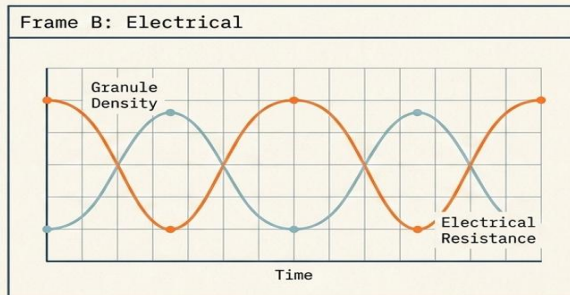
Hybrid - A balance transformer crucial for converting 2 line wire into 4 line wire.

The Transducer: Carbon Microphone Mechanics



The Transducer Principle

Carbon granules conduct electricity. Their electrical resistance is directly proportional to how densely they are packed together.



The Chain Reaction

Sound waves impinge on the diaphragm → Diaphragm vibrates → Granules compress and expand → Electrical resistance changes synchronously with the acoustic sound.

Definition

A **carbon microphone** is a transducer that converts **sound energy into electrical signals** using carbon granules.

Working Principle

1. Sound waves hit the **diaphragm**.
2. The diaphragm vibrates.
3. The movable electrode compresses and releases the **carbon granules**.
4. The density of the carbon granules changes.
5. As density changes, **electrical resistance changes**.
6. This produces a changing electrical current that carries the voice signal.

Chain Reaction (Slide Based)

Sound Waves → Diaphragm Vibrates → Carbon Granules Compress/Expand → Resistance Changes → Electrical Signal Produced

3. Why is a Small Amount of Sidetone Necessary? (5 Marks)

Definition

Sidetone is hearing a small part of your own voice in the receiver while talking.

Why is it Necessary?

1. It lets the caller know the phone is working.
2. It helps maintain a natural speaking level.
3. Without sidetone, people tend to **shout**.
4. Too much sidetone makes people **speak too softly**.
5. Therefore, a **balanced (small) amount** of sidetone gives the best communication.

4. Low & High Sidetone (Anti-Sidetone Circuit) (5 Marks)

Low Sidetone (Transmitting)

When we speak,

- Current in **L1 and L2 flows in opposite directions**.
- Magnetic fields cancel each other.
- Very little current is induced in **L3 (Receiver)**.
- We hear only a small amount of our own voice.

High Sidetone (Receiving)

When another person speaks,

- Current in **L1 and L2 flows in the same direction**.
- Magnetic fields add together.
- Strong current is induced in **L3**.
- Incoming voice is heard clearly.

Anti-Sidetone Circuit — Transmit vs. Receive

Transmitter — currents oppose → low sidetone

Receiver (L3)

Speech (transmit)

Receiver — current adds → high sidetone

Receiver (L3)

Incoming signal

Transmitting — low sidetone
Current through L1 and L2 flows in **opposite directions**, so only a small resultant current is induced in L3. The speaker hears little of their own voice.

Receiving — high sidetone
Current through the coils flows in the **same direction**; the resultant current adds and is induced in L3 — so the incoming caller is heard clearly.

Sidetone

Definition:
Sidetone is the **speaker's own voice** heard in the receiver (earpiece) while talking.

Echo

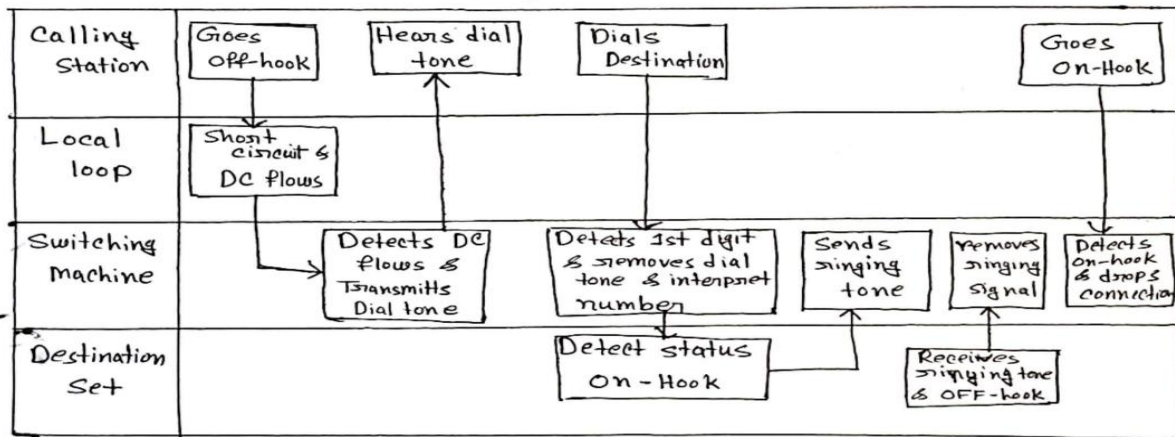
Definition:
Echo is the **reflected voice** that returns to the speaker due to **impedance mismatch** in the transmission path

COMPARISON

Sidetone vs. Echo

 SIDETONE	 ECHO
Sidetone	Echo
The speaker's own voice fed back to their own earphone.	Audio reflected back to the transmitter due to impedance mismatch.
A small amount is necessary.	Always unnecessary and annoying.
Anti-sidetone circuits reduce undesired sidetone.	Echo suppressors / cancellers are used to overcome it.
Too much makes users lower their volume.	Too much echo is highly distracting.

Telephone lifecycle



How a Telephone Call Is Made



- 1 Calling station goes off-hook.
- 2 Machine detects DC continuity and returns dial tone.
- 3 Caller dials the destination number.
- 4 On the first digit, dial tone is removed.
- 5 Number is interpreted; the destination loop is seized.

- 6 Before ringing, it checks if the far set is on- or off-hook.
- 7 Off-hook → busy signal; on-hook → ringing signal sent.
- 8 Destination answers; the loop completes and DC flows.
- 9 Switching machine removes the ringing signal.
- 10 Either end hangs up → the connection is dropped.

Functions of Telephone -

1. Notify the device for incoming calls.
2. Provides signal when call is accepted or acknowledged.
3. Converts speech signal to Electrical signal.
4. Regulates the amplitude of speech signal.
5. Provides ON/OFF hook features.

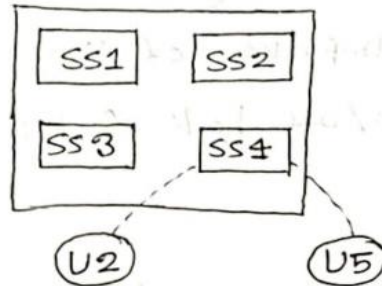
available Not available.

What is switching system?

It is a system, that coordinates the routing of communication.

Ex-

Between user 2 & 5.



Here, U2 is connected to U5 by switching system.

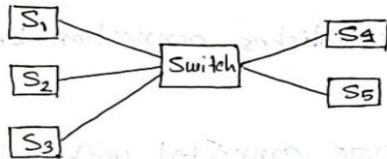
Why is a Switching System Important? (5 Marks)

1. **Establishes Connection** – Connects the caller with the receiver.
2. **Routes Calls** – Selects the correct path for the call.
3. **Efficient Resource Use** – Shares network resources efficiently.
4. **Supports Many Calls** – Allows many users to communicate at the same time.
5. **Disconnects Calls** – Releases the connection after the call ends so it can be used again.

=>



In point to point link,
total link = $\frac{n(n-1)}{2} = \frac{5(5-1)}{2} = 10$



But in switching system
for 5 user, it need 5
connection links.

Caller ID (Easy 2–3 Marks)

Definition:

Caller ID (CID) is a telephone service that **shows the caller's information (name and/or phone number) on the receiver's phone before it rings.**

Easy Definition:

Caller ID is a service that displays the caller's phone number or name before the phone rings.

How Caller ID Works (Easy 5 Marks)

1. The exchange detects an incoming call.
2. It collects the caller's information (name/number).
3. The information is sent using FSK (Frequency Shift Keying).
4. The receiver's phone decodes the data.
5. The phone displays the caller's details before ringing.

SDMF (Single Data Message Format) –

message type → message length → Date → Time → Number → Checksum

- Easy implementation
- Only number appears.
- Faster transmission
- Can't add new info easily.
- Suitable for older telephone.

MDMF (Multi Data Message Format) –

message type → message length → Parameter 1 → Parameter 2 → Parameter 3 → Checksum

- Shows caller name

- Future expansion

□ Traffic handling capacity

$$TC = \frac{2 \times SC}{N}$$

SC = Switching capacity.

N = Number of subscriber.

Ex: For a telephone line, Switching capacity = 40, No. of subscribers = 100
Find Traffic handling capacity.

$$\frac{2 \times 40}{100} = 0.8$$

Importance

□ EUF (Equipment Utilization Factor)

$$EUF = \frac{\text{Switching Elements in operation}}{\text{Total available element.}}$$

Ex: Active switching element = 80

Total SE = 100

$$EUF = \frac{80}{100} = 0.8 \text{ (80\%)}$$

□ CCI (Cost Capacity Index)

$$CCI = \frac{N \times SC}{C}$$

Ex: For a telephone exchange, No. of subscribers = 100
Switching capacity = 40
Cost = 20,000 TK

Find CCI = ?

$$CCI = \frac{100 \times 40}{20,000} = 0.8 \text{ (80\%)}$$

For cost CCI

For traffic TC

□ Which telephone is better-

TC $\Rightarrow$ If $TC_1 > TC_2$ then TC_1 is better.

Cost $\Rightarrow$ if $CCI_1 > CCI_2$ then CCI_1 is better.

Decode the message -

$\begin{array}{cccccccccccc} \underline{30} & \underline{20} & \underline{01} & \underline{08} & \underline{30} & \underline{33} & \underline{32} & \underline{39} & \underline{30} & \underline{32} & \underline{07} \\ \underline{08} & \underline{4A} & \underline{4F} & \underline{48} & \underline{4E} & \underline{20} & \underline{44} & \underline{4F} & \underline{45} & \underline{02} & \underline{0A} \\ \underline{38} & \underline{30} & \underline{30} & \underline{35} & \underline{35} & \underline{35} & \underline{31} & \underline{31} & \underline{31} & \underline{32} & \underline{7D} \end{array}$

→

Type = MDMF (30)

Length = Hex 20 → Decimal 32

Date & Time

Type = 01 (Date) & Time

Length = Hex 08 → Decimal 8

$\begin{array}{cccc} \underline{30} & \underline{33} & \underline{32} & \underline{39} \\ \underline{0} & \underline{3} & \underline{2} & \underline{9} \end{array}$

$\begin{array}{cccc} \underline{30} & \underline{39} & \underline{30} & \underline{32} \\ \underline{0} & \underline{9} & \underline{0} & \underline{2} \end{array}$

month Day Hour minute

Month = 03 (March)

Day = 29

Hour = 09 AM

Minute = 02

∴ March 29, 09:02 AM

Name

Type = 07 (Name)

Length = Hex 08 (Hex) → Decimal 8

$\begin{array}{cccccc} \underline{4A} & \underline{4F} & \underline{48} & \underline{4E} & \underline{20} & \underline{44} & \underline{4F} & \underline{45} \\ \underline{J} & \underline{O} & \underline{H} & \underline{N} & \underline{-} & \underline{D} & \underline{O} & \underline{E} \end{array}$

↳ space

∴ Name = JOHN DOE

Number

Type = 02 (Number)

Length = Hex 0A → Decimal 10

$\begin{array}{cccccccc} \underline{38} & \underline{30} & \underline{30} & \underline{35} & \underline{35} & \underline{35} & \underline{31} & \underline{32} & \underline{31} & \underline{32} \\ \underline{8} & \underline{0} & \underline{0} & \underline{5} & \underline{5} & \underline{5} & \underline{1} & \underline{2} & \underline{1} & \underline{2} \end{array}$

Number = 8005551212

So result,

Date
Month = March 29

Time = 09:02 AM

Name = JOHN DOE

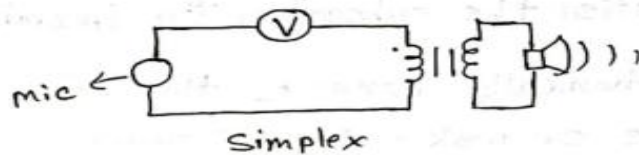
Number = 8005551212

Simplex vs Half duplex -

=>

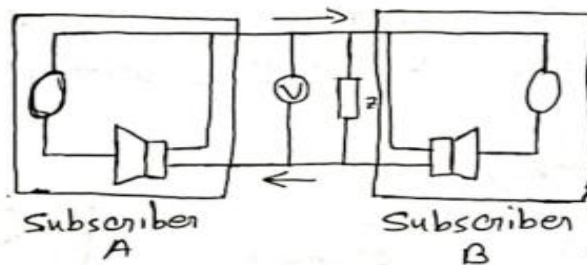
Simplex -

- It is a one way communication. The simplex circuit has an isolated transmitter and receiver loop, allowing it to communicate one direction.



Half-Duplex -

- It is a two way communication method but not simultaneous. It has a shared circuit.



Folded network -

- A symmetrical network in which the number of inlets are equal to the number of outlets.

